# KERALA WATER AUTHORITY

## Jal Jeevan Mission (JJM)-JJM-WSS to Munnar, Pallivasal, Devikulam & Chinnakanal Panchayaths in Idukki District- Design and Construction of 22 MLD Treatment Plant near Mattupetti Dam - General Civil Works

**E -TENDER No: 14/2024-25/SE/PHC/IDK/7346**

**DUE DATE : 29/11/2024** at 4.00PM.

## VOLUME–II

# SCOPE OF WORK

**OFFICE OF THE SUPERINTENDING ENGINEER**
**P.H. CIRCLE, KERALAWATER AUTHORITY**
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Tenderer

Superintending Engineer

**NAME OF WORK: - JAL JEEVAN MISSION (JJM) WSS TO MUNNAR, PALLIVASAL, DEVIKULAM & CHINNAKANAL PANCHAYATHS IN IDUKKI DISTRICTS – DESIGN AND CONSTRUCTION OF 22 MLD WATER TREATMENT PLANT**

## 1. GENERAL INFORMATION

The Project is aimed for providing potable drinking water to the entire area of Munnar, Pallivasal, Devikulam & Chinnakanal Panchayaths in Devikulam & Udumbanchola Taluks in Idukki District of Kerala State. The scheme area is situated at the Central North part of Idukki district of Kerala. Munnar panchayath are surrounded by Pallivasal, Mankulam, Marayoor, Kanthalloor, Vattavada, Devikulam, Chinnakanal & Bisonvalley panchayaths and Pallivasal panchayath are surrounded by Munnar, Bisonvalley, Velathooval, Adimali & Mankulam panchayath. Devikulam panchayath are surrounded by Chinnakanal, Munnar, Marayoor, Kanthalloor & Vattavada panchayath. Chinnakanal panchayath are surrounded by Devikulam, Bisonvally, Rajakumary & Santhanpara panchayath.

Design period of the scheme is taken as 30 years from 2023 A.D. with a per capita demand of 100 lpcd.

The scope of work includes Construction, trial running, commissioning and maintaining the major components of the water supply scheme for 18 months after commissioning and handing over to Kerala Water Authority. The detailed design and drawing should be submitted within 30 days from the date of agreement.

The major components are construction of 22 MLD WTP near Mattupetti Dam in Devikulam Panchayath in Idukki district of Kerala State.

## 2. CONCISE SCOPE OF WORK

This tender is invited for **Design, Construction, commissioning and maintenance of 22 MLD output capacity Water Treatment Plant.**

## 3. DETAILED SCOPE OF WORK

**3.1 Design, Construction, commissioning and maintenance of 22 MLD output capacity Water Treatment Plant.**

3.1.1 The Scope of the tender covers the

(a) Design, Construction, commissioning and maintenance for a period of 18 months 22 MLD output capacity Water Treatment Plant completed in all respects including construction of the structure for the various units, including clear water sump for 1.5-hour storage, supply and installation of chlorinator and all other connected works, including commissioning of the plant. Design should be made by utilizing existing free lands and with minimum demolishing of existing structures.

(b) Raw water will be delivered at the intake point of the aerator. From Aerator of the treatment works by the Authority and all works beyond this point for treatment of the raw water to conform to the drinking water standards are included in the scope of the work. Scope of work includes construction of a clear water sump of **14.00 lakhs** litter at the Treatment Plant site near Mattupetti Dam in Devikulam Panchayath. The construction includes all RCC structures including inlet, outlet, over flow and scour arrangements, landscaping and gardening, compound wall all around the site, approach road to plant site, internal roads, all necessary instrumentation electromechanical works etc. complete.

3.1.2 The Treatment Plant, reservoirs and all other installation shall be based on sound engineering practices and techniques in Water Treatment consistent with the requirement of National building code, BIS, CPHEEO Manual on Water Supply and treatment (latest edition), economy and in accordance with the specification provided in this document. The layout of WTP and detailed Design and drawings of all componentsof WTP shall be

submitted and got approved from the agreement authority before execution of the work without any extra cost.

3.1.3 It is to be understood that all works connected with the construction and installation of the water treatment plant in complete shape for its proper and satisfactory functioning have to be included by the tenderers in the tender even if each of the components are not explicitly stated in this schedule. The main works required are listed below: -

**A. CIVIL WORKS**

The civil work of treatment plant shall include:

i. Design and Submission of Detailed drawings of 22 MLD Treatment Plant with All Components like Aerator, Flash Mixers, Flocculators & Clarifiers, Raw water and Clear Water channels, Chemical house & Filter House, wash water tanks, Chlorinator room, Laboratory, Transformer rooms as per IS codes and Standards.

ii. **Aerator** - Construction of multiple tray cascade aerator unit Contains 4 trays and collection tray and Raw water channel (minimum 30 cm free board) with IS codes and standards. Provide suitable foundation, plastering, finishing, painting and tile works as per the Design and direction of engineer in charge.

iii. Construction of components in 22 MLD treatment plants (one unit of Flash mixer, one unit of Flocculator, two unit of Clarifier, filter media min 8 unit filter bed, wash water tank, two units of Alum tanks, two unit of lime tank and a clear water sump) as per IS codes and standards and also the direction of engineer in charge.

iv. Construction of **Chlorinator room** floor area of 97.13 $m^2$ as per specification of Factories & Boilers Department.

v. Construction of **Laboratory** with all facilities like storage area for chemicals and apparatus. There shall be a Stainless-steel sink of appropriate size and one stainless steel tap to supply water to the sink and provide required furniture arrangements for laboratory as per design.

vi. Construction of **Transformer Control rooms** floor area of 145 $m^2$ with 4 toilet attached rooms and one transformer room as per IS standards and to be accommodated above the sump.

vii. Construction of **Clear water pump house** shall have minimum height of 4.00 m and total floor area of 145 $m^2$ to accommodate 6 sets of centrifugal pump sets of various capacities. Pump house is to be constructed to accommodate pump sets, Panel board, starter and other accessories so as to facilitate easy inlet and outlet connection of pump sets shall be provided.

viii. Construction of **Clear Water Reservoir** of capacity **14 lakhs liters**. Top level of the Clear Water Reservoir shall be +1578.179m.

ix. All closed or open channels or conduits connecting various units, with all details of pipes, flanged pipes, specials, fitting etc. specifically in the case of under drainage system of the filter unit.

x. All drains and manholes for collection and disposal of sludge from the site for final disposal to the nearest water source.

xi. **Scour** arrangements of all units shall be above or at maximum flood level.

xii. Necessary by pass arrangement for allowing (i) the raw water after aeration into the clear water sump (ii) for allowing raw water after aeration direct into the filter bed and

(iii) for allowing clarified water direct into the clear water sump. In all cases access to chlorination should be provided.

xiii. **Raw Water inlet** arrangements shall be of a permanent structure with shutters and screens and control valves.

xiv. Lightening arrestors wherever necessary as per ISS and as per other relevant rules should be provided.

xv. Calibrated notches for flow measurement shall be provided at the inlet to the plant andin the clear water channel of the plant for manually measuring the quantity.

xvi. **Compound wall** around the treatment plant yard with R.R Masonry foundation & basement, superstructure with wire cut brickwork/Solid Concrete blocks in C.M.1:6 plastered and painted as per specification. Suitable strong and durable fencing materials such as chain links or other such materials fixed in suitable M.S angle iron sections shall be provided above the brick wall.

xvii. Construction of **Office room, toilet, Lobby** and **car porch** as per the size in design and IS codes and Standards.

xviii. Raw water inlet channel and partial flume

xix. Inlet well connecting to the clarifloculators

xx. Clear water leading channel from Filter house up to the clear water sump

xxi. 3 nos. **Toilet rooms** inside the Filter house with **21m$^2$** area minimum

xxii. Measures required for protecting structures against earth quakes

xxiii. **Blower room** **34.5 m$^2$** area minimum

xxiv. **Foundation** is raft foundation in Proposed Site.

xxv. Design should contain minimum destructions to the existing structures and Clearing jungle including uprooting of rank vegetation, grass, brush wood, trees and removal of rubbishes. Cutting trees, conveying, stacking, etc..

xxvi. Preparation of site including leveling to finished yard level, laying concrete paving tiles (approved by Authority) not less than 5 M around the structures.

xxvii. **Bridge for supporting pipe laying** – It includes Design, Construction of RCC bridge for supporting pipe line across the spillway. The following components shall be done in this scope

1.foundation, 2. column up to plinth, 3. plinth beam & 4. Deck slab of width 2m etc, as per specification.

xxviii. Internal roads of 100 m formation with 4 m wide concrete and concrete paver block 540 $m^2$ surface connecting all components.

**B. MECHANICAL WORKS**

The mechanical works shall include

i. Supply and erection of all mechanical equipment's required for the raw water measuring devices, integrations and automatic recording at raw water channel.

ii. Supply and erection of necessary mechanical equipment's for dosing of the chemicals or others solutions in the chemical house.

iii. Supply and erection of necessary mechanical equipment's for mixing of chemical in the mixing chamber. One set mixing equipment's with motors to be supplied as stand by for lime and alum tanks.

iv. Supply and erection of necessary mechanical equipment's for flash mixtures and floculators.

v. Supply and erection of necessary mechanical equipment's for clarifier including constant bleed arrangements to monitor the concentration of sludge.

vi. Supply and erection of all mechanical equipment's required for the under drainage of the filters, washing of filters, filter rate controllers, filter flow gauges, loss of head indicators, wash water controllers and indicators, filter media with all necessary pipes and valves etc. of approved quality.

vii. Supply and erection of two Gravity type chlorinators of adequate capacity, with all mechanical equipment's in the chlorinator room, one working and the other as stand bye. The chlorinator should be attached with Barrow metric leg arrangement. The materials of pipe should be copper. Each chlorinator has a capacity for disinfection of treated water under 100% over loading condition.

viii. Supply and erection of all mechanical equipment's for proper washing of filter includingair blowers/ compressors, motor and pump and accessories for lifting the water to the wash water tank. Standby to be provided for blowers & pumps and accessories.

ix. Supply and erection of all mechanical equipment's for the wash water tank such as inlet, outlet and over flow pipes and control valves. Water level gauges, over flow alarm devices, ventilators, scour arrangements, ladders for access to top of tank, manholes and ladders with side railings for access to the inside of the tank etc. including pumping arrangements.

x. Supply and erection of all mechanical equipment's for the clear water reservoir for its proper functioning such as inlet, outlet and overflow and scour arrangements with

necessary pipes and control valves, water level indicators, manholes, ladders, ventilating arrangements etc.

xi. Supply of 2 nos 1000kg chlorine turners and 5 Nos. filled 100 Kg chlorine cylinders (with appropriate certificates in original) for drawal of chlorine required for disinfection of the filtered water. The supply shall also include necessary safety equipment's and protection devices for handling chlorine.

xii. Supply and erection of all scientific equipment such as turbidity meter, P.H. meter, Residual Chlorine test kit, jar test apparatus, field test kit, glass ware etc. and mechanical equipment's for the proper functioning of the laboratory and also supply of the chemicals and other consumable for analysis during the maintenance period.

xiii. Supply and erection of mono rail or any such appropriate devices for the movement in all direction in chlorine room for handling cylinders.

xiv. Lifting devices for chemicals.

xv. Platform weighing balance for chemicals.

xvi. Supply Suitable lifting devices (2 nos Gantry crane arrangements 3.0 T capacity for facilitating the lifting & erection of pump sets in the clear water pump house and lifting and placing in position of chlorine turners at chlorine room each).

xvii. Dial type indicator shall be provided at easily accessible points for indicating rate of flow of air and water used for back washing purposes.

xviii. Safety equipment's & devices may be provided for handling chlorine.

## C. ELECTRICAL AND ELECTRO MECHANICAL WORKS

The electrical and electro mechanical works shall include

i. Supply and erection including necessary wiring of all electrical equipments like electricmotors etc., accessories like starters, switches, meters, switch boards etc. and erection and other appurtenances such as panel board with TP switch and fuse units of adequate capacity for receiving KSEB supply and all installations for the satisfactory operations of the plant. All such electrical works shall be in conformity with IE rules prevalent.

ii. Providing necessary lighting arrangements at the Treatment plant and Clear Water sump sites so that sufficient light available during night hours for the normal working of all these facilities as if during day including cost of all materials, working charges etc. complete

iii. Preparation of necessary documents through authorized electrical contractors for submission to KSEB for obtaining power connection to the plant shall also form part of the work to be carried out.

iv. It is the responsibility of the Firm/Tenderer to observe all IE rules, get sanction for all electrical works. Power connection shall be obtained on behalf of KWA, Statutory fees in the account of power connection to be borne by the contractor then and there and same will be reimbursed by KWA on production of vouchers.

v. The accessories such as Panel board change over switches, capacitors cables earthing (as per IE rules) and all other related accessories should confirm to KSEI specification.

## D. WATER SUPPLY AND SANITATION ARRANGEMENTS.

i. Water supply and sanitary installations to the building including three toilets and a septic tank are to be provided, attached with the Filter house. Necessary sinks and taps required at the laboratory shall also be provided as part of the work.

## E. REQUIREMENTS FOR VARIOUS COMPONENTS OF PLANT

### 3.2.1 General conditions:

The registered contractors/ firms who are not the manufactures/ authorized suppliers of the Mechanical and Electrical equipment's for the water treatment plant should furnish the names of the manufacturers/ authorized suppliers from whom they intend to procure the mechanical and electrical equipment's required for the plant. Testimonials from the manufactures/ suppliers showing their previous experience in the supply, erection and maintenance of similar kinds of works, an agreement signed by the authorized signatory from the supplier/ manufacturer supporting the contractor/ firm guaranteeing the supply, erection and maintenance of such equipment's and also a letter addressed to the Superintending Engineer, Kerala Water Authority, PH circle, Idukki by the suppliers/ manufacturers to the above effect shall be provided with technical bid. A list of plants where such equipment's have already been provided/ installed with the year of such installation and full address of the officer managing the plant shall also be provided for reference.

3.2.1.1 The physical, chemical and bacteriological analysis report of a few samples of raw water is appended. The turbidity of raw water may go up to 100 NTU or even higher. The

treatment process shall be an efficient and economical one so as to obtain the effluent conforming to the drinking water standards of treated water as per CPHEEO Manual. It shall be the tenderer's responsibility to produce drinking water as per the standards given by suitably designing and constructing the treatment plant. The technical aspects of the treatment plant offered by the tenderer and the details of the units shall be elaborated in separate sheets and attached with the tender.

**3.2.2 Aerator:**

Aerator shall be multiple tray/cascade type with minimum 4 trays and a collecting tray. Aerator trays shall be constructed in M 25 R.C.C. Aerator Shall be designed for 22 mld intake. The diameter of inflow pipe shall be in suitable size and diameter of collection tray.

**3.2.3 Measurement of Raw Water Flow:**

There shall be adequate arrangements for the automatic recording of the flow of raw water. There shall be a direct reading dial type meter with integrators etc. located in the operating room. The meter, if power operated shall automatically switch on to battery operation in case of power failures. A notch of suitable shape and size shall be provided in the raw water channel to measure manually the quantity of inflow to the plant and outflow from the plant. A suitable calibrated scale shall be provided near the notch to know the quantity of flow.

**3.2.4 Chemical House**

3.2.4.1 Adequate facilities shall be provided for the storage of chemicals required for the treatment of water for a period of two months, calculating the requirement on the basis of conditions of raw water. Facilities for weighing the chemical (digital model) before feeding

into the solution tanks and convenient means for lifting the chemicals from the stores to the

Alum and lime solution tanks shall be provided. For storing chemicals, the floor area should be a minimum of 191.5 $m^2$ with separate area marked for each chemical. Flooring of chemical house shall be with 100mm thick CC M20 & finished with Matt finish floor tiles.

3.2.4.2 Solution feeding shall be adopted. There should be at least two tanks for each chemical feed. The capacity of each tank shall be such as to hold 8 hours requirements at the maximum demand of chemical for the design flow. Dissolving trays or boxes and also facilities for draining the solution tanks should be provided. The solution tanks shall be located as near as possible to chemical storage go down. These tanks shall be located at a suitable elevation to facilitate gravity feed of the chemical's solution. A mechanical lifting arrangement for lifting the chemicals to the solution tanks shall be provided. Arrangements for continuous mechanical agitation of the solution in the tank shall be provided. All the pipeline, valves, specials etc. in contact with chemicals shall be made with corrosion resistant materials. Necessary solution level indicators shall be provided for the tanks. The solution feeding equipment shall be capable of automatically adjusting the quantity of solution feed according to the quantity and quality of raw water flow. Also the feeding equipment shall be capable of feeding the solution at desired rates continuously and accurately irrespective of the level of the solution inside the tanks.

**3.2.4.3 Porch**

There shall be a car porch of minimum area 18$m^2$ in front of the lobby which shall be connected to the road. But at a minimum elevation of 15 cm from the road. The foundation shall be provide with suitable site conditions.

**3.2.5 Flash Mixer-Rapid Mixing**

There shall be One flash mixer. A minimum detention time of 60 seconds should be provided. Necessary mechanical equipment for rapid mixing shall be provided. Isolation of the flash mixers from aerator flocculator channel shall be made possible by suitable watertight sluice gates.

**3.2.6 Floculator -Slow Mixing**

There shall be 2 nos flocculators. Necessary mechanical equipment for slow mixing may be provided so as to have a minimum detention time of 30 minutes.

**3.2.7 Clarifiers**

There shall be 2 nos clarifiers of minimum dia. **22m** providing a minimum detention time of 2.5 hrs. Necessary mechanical equipment should be provided. The clarifier shouldbe capable of giving clarified water having turbidity not exceeding 10 NTU. Combining flocculation and clarification in two structural unit are also acceptable if found cheaper. The surface overflow rate should be between 30 to 40 $m^3/m^2$/day. There shall be an appropriate walkway around the clarification system with hand rails. An extra scour arrangement should be provided in addition to normal sludge pit with necessary valve control. Constant bleed arrangements shall be provided. The peripheral launder wall should preferably be provided/furnished with fish plate of corrosion resistant material. Construction of granular sub- base by Providing close graded Material conforming to specifications, mixing in a mechanical mix plant at OMC, Carriage of mixed material by tippers to work site, for all leads & lifts, spreading in uniform layers of specified thickness with motor grader on prepared surface and compacting with vibratory power roller to achieve the desired density, complete as per specifications and directions of Engineer-in-Charge. With material conforming to Grade - III ( size range 26.5 mm to 0.075 mm) having CBR Value – 20.

**3.2.8 Rapid Sand Gravity Filters.**

3.2.8.1 The filters shall be designed based on normal rate of filtration of 80 lpm/m2. But the inlet and outlet control arrangements shall be designed to permit 100% over load for emergency situations.

3.2.8.2 There shall be a minimum of 8 Nos. of independent filter beds.

3.2.8.3 The filter chambers and cross gutters shall be of R.C.C. M 30 and side walls shall be lined with ceramic glazed wall tiles of approved quality from top of filter walls to 300 mm below the top of sand bed. Walk way on the side shall be provided around and in between filter beds with suitable handrails in stainless steel pipes. The minimum width of walkway shall be 900 mm.

3.2.8.4 The filter media shall be sand supported by gravel and shall be as per relevant IS specification. Before the actual loading of filter media into filter chambers, the chemical & Physical properties of the proposed media materials shall be got tested at approved laboratories and accepted by the Executive Engineer/Agreement Authority.

3.2.8.5 A suitable type washing system including air wash, which gives an efficient washingof filters shall be provided. The wash water consumption should not exceed 2% of the quantity of water filtered in between washing and the duration of wash shall not normally exceed 10 minutes (7 minutes where air and water is used). The water standing on the filter bed at the close of the wash shall be clear with turbidity not exceeding 10 NTU.

3.2.8.6 The filter runs should normally be not less than 24 hours with a loss of head not exceeding 2m.

3.2.8.7 Rate of flow and loss of head gauges shall be provided for each filter. The filter valves may be hand operated and located in such position as will facilitate easy operation. A filter rate controller shall be provided for each filter.

3.2.8.8 The filters and appurtenances shall be roofed over.

3.2.8.9. The following velocities may be allowed in designing the various pipes in the pipegallery.

| | |
|---|---|
| Influent | : 0.30 to 0.6 m/sec. |
| Effluent | : 1.2 m/Sec. |
| Wash water | : 3 m/ Sec. |
| Waste Water | : 1.5 - 3.0 m/sec. |
| Wash air | : 12.8m/sec. |

All pipes in the pipe gallery shall be of CI/DI with suitable class or equivalent type, unless specified otherwise. If fabricated specials are found inevitable, such fabrications shall be done with M.S. sheets of thickness not less than 10 mm and the welding procedure shall be got approved from the Executive Engineer. All fabricated pipes/specials shall be painted with good grade epoxy paint inside after cleaning the surface thoroughly & over an epoxy priming coat. Outside painting shall be with anticorrosive black paint.

3.2.8.10 The under-drainage system shall be so designed and properly constructed so that no damage is sustained to the under-drainage system or to the filter beds even in case of any increased air/water pressure. 10% of the quantity of total No. of perforated pipes/ specials/ nozzles required shall be supplied as extra for maintenance purpose. The under-drainage system may be of MDPE or class 8 PVC or higher grade. The under- drainage system proposed shall be specified in the Technical Bid.

3.2.8.11 The Supply of wash water shall be made through an overhead storage tank capable of washing 8 filter beds at a time at the rate of 80 lpm/m2 for 7 minutes and may be provided over filter house. The work shall include all pipe connections to and from the

wash water tank, all mechanical equipment like wash water pump, motor, etc. An electrically operated level indicator working in conjunction with a float device shall be provided to show the water level in the wash water tank. The control for the wash water pumps and other mechanical devices shall be installed in the operation room inside the filter plant. The under-drainage system shall be designed for a back wash rate of 80 lpm.

3.2.8.12 Compressor should be provided capable of discharging compressed air at the rate of $0.9m^3$ /$m^2$ /min with 100% stand bye. Blower room of minimum area of 34.5 m2 to be provided.

3.2.8.13 The wash water pump set shall have a minimum capacity to fill the wash water tank in 2 hrs against a total head of 17 m. The pumps shall be of reputed make coupled and split casing with slow speed motor. The foundation shall be of raft foundation.

**3.2.9 Disinfection**

3.2.9.1 For disinfection of water, chlorination has to be done. The work shall include supplying and installing Gravity type feeding chlorinator for feeding chlorine from chlorine cylinders. The equipment shall have provision for regulating the dosage as well as indicating the dosage.

The capacity of the chlorinators shall be such as to have effective disinfection while the filters working at 100% overload. Provision for feeding bleaching powder solution at the time of exigencies such as gas and chlorines devices goes out of order may be provided.

3.2.9.2 One stand by chlorinating unit and spares required for periodical maintenance for a minimum period of 18 months shall be provided.

3.2.9.3 There shall be a separate room with minimum area of 97.13 $m^2$ for housing the chlorinator with necessary ventilating arrangements and sufficient moving space for the operators. The chlorine room shall be easily accessible close to the point of application and convenient for truck loading and safe handling of chlorine cylinders. The stand by chlorinator shall be filled along with the main system.

3.2.9.4 2 nos 1000kg chlorine turners and 5 Nos. filled 100 Kg chlorine cylinders are to be supplied with necessary safety certificates. The cylinders should be filled from T.C.C. Aluva or any such approved manufacturers of chlorine.

3.2.9.5 There shall be a hand operated easy moving device (crane of 3.0T capacity) for handling turner/cylinders. All other provision in the chlorine room as well as its construction shall strictly conform to the well accepted safety norms and relevant code of practices. Roof of chlorine room shall be of mix M25 and clear cover for the steel reinforcement shall not be less than 40 mm for slab and 50 mm for beams. Area of chlorine room shall be minimum 97.13 $m^2$, standby arrangements to disinfect with bleaching powder solution at times of exigencies should also be provided. Space for storing chlorine cylinder shall also be provided.

3.2.9.6 Essential safety instructions for handling chlorine gas including precautions in case of chlorine gas leakage shall be written in a 1.00 m x 0.90 m size board & exhibited at chlorine room entrance. Gas mask must be available near the chlorine room and at the office room.

**3.2.10 Clear Water Sump**

3.2.10.1 The capacity of the sump shall be 14 lakhs liters. The foundation shall be suitable

with the soil condition of the site.

3.2.10.2 Arrangements for cleaning and scouring the sump may be provided. Arrangements for scour and overflow including vertical connection shall be provided for the overhead back washing tank also. Water level indicators showing the water level in the sump shall be provided so as to read from the inside of the control room. The sump shall be rectangular or circular in shape.

The site shall be leveled properly. MS ladder shall be provided for entering into the sump. Design shall be in accordance with IS 456:2000, IS 3370 (Part II) – 2009 with latest amendments. The design shall be got approved by the agreement authority before construction work is started.

1. A minimum free board of 30 cm below of the cover slab / beam shall be provided.
2. Soil investigation to assess the bearing capacity of soil shall be done by the contractor through reputed agencies at his own cost for designing the structures.
3. The access to the tank sites shall be formed suitably cutting or filling earth as per site requirements.

Any further requirements related to the construction work if directed by concerned engineers have to be met with. All technical specifications has to be fully satisfied. All the design shall be in accordance with IS 456:2000, IS 3370 (Part II) - 1991 with latest amendments. The design shall be got approved by the agreement authority before construction work starting.

3.2.10.3 As a matter of aesthetics, the chlorinator, pump house and transformer control room must be constructed on the top of sump with required number of window

door partitions ethnicities.

### 3.2.11 LABORATORY

3.2.11.1 A fully equipped laboratory of area not less than **78.06 $m^2$** for conducting routine test for turbidity, pH, Residual chlorine, dosing of chemicals etc. shall be provided. One numbers of jar test apparatus shall be provided. Test reagents required for 6 months use shall also be supplied. The make specification of the equipment shall be provided in the technical bid. The work bench shall be minimum 75cm wide and covered on top with granite slab & sides with Ceramic tiles of good quality. There shall be a Stainless-steel sink of appropriate size and one stainless steel tap to supply water to the sink.

There shall be 2 no office table with drawer on both ends, 6 nos of armed chair, 1 no steel Almirah with locker of size 1.80mx0.90mx0.45m,4 nos of stools (teak wood), 1 nos of glass door cupboard, table, chair& coat 1 no each, 4 nos of ordinary chair of reputed make or nearest size inside the lab. All furniture shall be of such reputed make as approved by the Engineer-in-charge. **List of all laboratory equipment are attached as Appendix.**

3.2.11.2. There shall be a **private room** of floor area not less than **25.66 $m^2$** with **toilet** facilities attached.

3.2.11.3. There shall be an **office room** of floor area not less than **35.15 $m^2$ with toilet** facilities attached.

3.2.11.4. There shall be a **lobby** of floor area not less than **37.15 $m^2$** in the ground floor

### 3.2.12 GENERAL LEVELS FOR DESIGN OF PLANT

3.2.12.1 The inlet level of the aerator shall be approximately +1585.00. The contractor shall fix various units of the plant approximately. The contractor can change the top level of

Aerator slightly according to his design of treatment plant but the pressure drop in each unit shall be according to CPEHO manual and there will be gravity flow till the sump from aerator. The change in the level will change the design of raw water and clear water pumps and hence while quoting the tenderer shall take into consideration all these factors. However, the Authority reserves the right to change the level and the level fixed will be final and binding. The foundation shall be raft foundation.

3.2.12.2 Bottom slab level of clear water reservoir shall not be below +1575.179 m approximately

3.2.12.3 In between these two levels the contractor shall fix various units of the plant appropriately. The ground level of the site for the plant is generally level with all classes of soil. Earth filling for leveling the site with contractors earth is to be done after construction of the structures.

**3.3. SEPTIC TANK**

Supply, installation and commissioning of one PVC/Concrete septic tanks (50-75 flushes per day) and interconnection with manhole including cost of pipes specials, labour.

**3.4. APPROACH ROAD TO PLANT.**

Approach road concrete surface 5m wide up to the Entrance gate of Plant shall be constructed from main road.

**3.5. ARCHITECTURAL FEATURES.**

3.5.1 The layout of the plant should be spacious, convenient and appealing. The entrance lobby shall have minimum 37.15 $m^2$ area. There shall be a porch of minimum area 18$m^2$ at the front entrance.

3.5.2 The architectural features of the plant shall be attractive. The structures should present a pleasing appearance with aesthetic features both inside and outside.

3.5.3 The entire floor inside the filter house including laboratory etc. must be finished with scratch free vitrified flooring of approval quality and colour. All hand railings inside the filter house shall be of stainless steel. The entrance lobby and stairs shall be provided with good quality scratch free vitrified finish. Anodized aluminum doors and windows may be provided with glazed panel for the structure except for office room and private room where the doors shall be of paneled Teakwood. For window openings MS square grill shall be provided for protection.

3.5.4 White glazed tiles shall be provided inside the inspection chambers with glass top and inside lighting. White glazed tiles shall be provided in the filter chamber also up to 30 cm below top of filter media and also side of wash water gutters and floor shall be finished with scratch free vitrified flooring tiles.

3.5.5 All RCC structures shall be plastered smooth and evenly by cement plastering with cement motor 1:4 mix, the chemical house and chlorinator room floor be plastered with 1:3 cement mix with a neat flushing coat.

3.5.6 The interior of the buildings shall be provided with emulsion paint of approved quality and colour, three coats on the internal walls. The exterior walls should be finished with anti-fungal paints such as “Apex” or “Weather coat” of approved colour.

3.5.7 All steel and wooden materials and equipments shall be painted two coats with best quality enamel paint and of approved colour over a priming coat of zinc chromate primer. The door at office room and private room shall be given lacquer polish.

3.5.8 Flocculaters/clarifiers/clarifoculators and inlet channel shall have walkway of minimum 100 cm width with suitable handrails.

3.5.9 All step ladders shall be of M S Fabricated.

3.5.10 The front door shall be of minimum 2.00-meter width and with powder coated

anodized Aluminium frame and glass panels.

3.5.11 **Landscaping , Gardening & Site levelling after construction:-** *A Pleasing and attractive landscaping of 500m$^2$ , plant area shall also be done along with an "**attractive garden" of minimum 500m$^2$** , Supply and Planting Buffalo Grass of 500 m$^2$ and site levelling after construction of 750 m$^2$ then also proper yard levelling by spreading baby metal over the entire ground by the tenderer as per the direction of engineer in charge*. The WTP compound should be provided with 1703 m$^2$ paving tiles other than garden areas and appropriate sub grade wherever necessary connecting all units shall be constructed and finished along with cross drainage works from the entrance gate (4m width) and a cattle trap.

3.5.12 Walk way shall be provided in between and all around the clariflocculators and filter house. The width of all walk way shall be minimum 100 cms with 850 mm height handrails. The provisions of the balustrades and rails shall be such that they shall be pleasing in appearance and matching to its position in the structure.

3.5.13 **External Yard Lighting**

3.5.13.1 Lighting Switch Board (LSB): "250A TPN bus bars of sizes: Phase : 50.8 x 6.35 mm Al-flat Neutral : 50.8 x 6.35 mm Al-flat Earthing conductor : 25 x 3 mm copper 250 A TPN Isolator 1 No. as incomer, 32 A TPN SFU 4 Nos. as outgoings. Metering compartments comprising of voltmeter, ammeter, selector switches, indicating lamps etc. complete as per KEI Standards."

3.5.13.2 Lighting Distribution Boards (LDB- I & II): Supply, erectioon, testing and commissioning of MCB distribution board comprising to the following: 40 A TPN 4 pole MCB 1 No. as incomer with 40A 30 m A ELCB. 6/10 A SP MCB - 18 Nos. as outgoings. With tinned copper busbars/cables, interconnection, housed in a suitable metal enclosure as

per KEI standards fixed in open system.

3.5.13.3 Cabling: Supply, laying, testing and commissioning of the following size PVC/XLPE insulated, sheathed armoured Aluminium conductor cable (1100/11000 Volts grade) clamped on wall/ceiling with clamps and saddle clips partly laid in cable tray/partly laid in trench. 4 core 16 Sq.mm PVC/XLPE AL Cable (LSB to LDB).

3.5.13.4 Cabling: Supply, laying, testing and commissioning of the following size PVC/XLPE insulated, sheathed armoured Aluminium conductor cable (1100/11000 Volts grade) clamped on wall/ceiling with clamps and saddle clips partly laid in cable tray/partly laid in trench -3 core 6 Sq.mm PVC/XLPE AL Cable (LDB to Streat Lights).

3.5.13.5 Cabling: Supply of all materials and providing end termination of the following size PVC/XLPE insulated, sheathed armoured Aluminium conductor cable (1100 /11000volts grade) for giving power connections to Transformers, Panel Boards etc. complete as per standards. 4 core 16 Sq.mm PVC/XLPE AL Cable.

3.5.13.6 Cabling: Supply of all materials and providing end termination of the following size PVC/XLPE insulated, sheathed armoured Aluminium conductor cable (1100 /11000volts grade) for giving power connections to Transformers, Panel Boards etc. complete as per standards.-3 core 6 Sq.mm PVC/XLPE AL Cable.

3.5.13.7 External Lighting: Supplying, erection and testing steel tubular swaged poles as per IS 2713 (Part II) - 1980 of height 7.5M above ground level and 1.5M below ground level bottom 139.7 mm dia and 114.3 mm dia for middle and 88.9 mm dia for top including providing base plate providing concreting in foundation, construction of pillar accomodating fuse and switch housed in weather proof box including wiring to the light fitting with 2 runs of 3/20 PVC insulated copper conductor. Supplying and fixing a 4 way 30 A rated terminal block include the weather proof box. The pole shall be painted with one coat of primer and

two coats of enamel paint as required by the Engineer in charge. The street pole shall be as per the drawing enclosed.-Single Bracket Type.

3.5.13.8 External Lighting: Supplying, erection and testing steel tubular swaged poles as per IS 2713 (Part II) - 1980 of height 7.5M above ground level and 1.5M below ground level bottom 139.7 mm dia and 114.3 mm dia for middle and 88.9 mm dia for top including providing base plate providing concreting in foundation, construction of pillar accomodating fuse and switch housed in weather proof box including wiring to the light fitting with 2 runs of 3/20 PVC insulated copper conductor. Supplying and fixing a 4 way 30 A rated terminal block include the weather proof box. The pole shall be painted with one coat of primer and two coats of enamel paint as required by the Engineer in charge. The street pole shall be as per the drawing enclosed -Double Bracket Type.

3.5.13.9 External Lighting: Supplying and fixing street light fittings with all accessories and 100 watts LED lamp fixed with above poles.

3.5.13.10 External Lighting: Supplying and fixing light fittings over compount wall allround the WTP yard with all accessories and 20 watts LED lamp fixed with above poles.

3.5.13.11 Safety Items: Supply and providing safety items as per Kerala Electrical Inspectorate and relevant standards. 2nos 1800X900X12mm 11KV rubber mat 2sets 11KV Hand Gloves 6nos. 11KV Danger board 1set Fire bucket with stand 2sets First aid box 2sets Shock treatment chart 2sets Fire extinguisher 5kg.

**3.6 PAINTING/FINISHING OF MECHANICAL EQUIPMENTS.**

3.6.1 All steel works for under water operations such as for flash mixer, flocculators & clarifiers shall be painted with good grade epoxy paint over a coat of epoxy primer after sand blasting and cleaning the materials at site. The inspection of the Executive Engineer in

charge shall be sought for after sand blasting but before commencing the epoxy primer paint and the fact should be reported to the agreement executing authority.

3.6.2 All steel work above water level shall be painted with 2 coats of approved quality synthetic enamel over a coat of zinc chromate primer after thoroughly cleaning the surface under sand blasting.

3.6.3 All cast iron pipes shall be painted with 2 coats anti corrosive black Japan paint after cleaning the surface with wire brush & sand paper.

3.6.4 All electrical panel boards should be painted with anticorrosive epoxy paint.

**3.7 INFORMATION & CAUTION BOARDS.**

The salient features of the filter plant & its schematic diagram shall be neatly painted in a 1.50x 11.20 m board & exhibited in the entrance lobby. Necessary caution boards should be exhibited at the control room, Chlorine room and chemical room and a title board at the front gate.

**4. SUPPLY & ERECTION OF SUITABLE CENTRIFUGAL PUMP SETS**

**1.** Design, supply, erection, commissioning and maintenance of suitable centrifugal pump sets and accessories. Minimum 2 Nos. pump sets which include 100 % as standby shall be provided in the clear water pump house for each zone. The initial and final levels of the reservoirs to be verified before fixing the head of the pumpsets.

**2.** Duty characteristics of pumps to be supplied-

| Sl No | Description | Type of pipe |
|---|---|---|
| 1 | WTP to Kuttiyarvalley GLSR- Discharge 82 lps, Head 54 m | 350 mm DI-K9 |

| | | |
|---|---|---|
| 2 | WTP to Santose Colony GLSR - Discharge 10 lps, Head 212 m | 150mm GI |
| 3 | WTP to Munnar GLSR - Discharge 135 lps, Head 97m | 450 mm DI-K9 |

## Specification of Centrifugal Pump set and accessories

### 1. Pump

Supply, erection, testing and commissioning of KWA pre-qualified reputed make Centrifugal pump sets with Bronze/SS impeller, SS shaft and CI pump casing with suitable type base plate with coupling, coupling guard foundation bolts & nuts etc. complete suitable for coupling the pump and motor above the base plate etc. complete including providing suitable concrete foundation including cost of cement sand and metal etc. complete. The duty condition is as follows

Discharge - (separately attached)

Total head - (separately attached)

Suction head - Positive suction

Length of suction pipe- Nil

Speed- <1500 rpm

### 2. Motor

Supply, erection, testing and commissioning of KWA pre qualified reputed make horizontal solid shaft foot mounted TEFC squirrel cage induction motor suitable for the above pump working 3 phase 50 Hz AC supply, working voltage 415 V/3300V (select the suitable). The motor shall confirm to EEF1/IS 325 including providing suitable concrete foundation including cost of cement sand and metal etc. complete For 3300v motors digital motor protection relay shall be included in starter as per present IE rules. Flexible coupling shall be used for coupling of pumps and motors.

### 3. Starter

Supply, erection, testing and commissioning of suitable rating (Fully Automatic starter with air break contact for less than 100 HP and for above 100 HP FCMA starters to be used.) starter suitable for the above motor with over voltage and under voltage protection,

single phase preventer, over load relay with motor protection relay with main contactor and bypass contactor as per IE rules and code of practice etc. complete.

**4. Panel board**

Supply, erection, testing and commissioning of cubical type floor mounted MS fabricated

dust and vermin proof common control panel board consisting of 1 No. suitable capacity MCCB as incomer and 2 Nos of suitable capacity MCCB as outgoing (the two MCCBs are interlocking with each other) providing suitable size Aluminium bus bar inter connect the above MCCBs and fitted with 3 Nos of indicator lamps, 1 No. volt meter with selector switch, 1 No. Ammeter with selector switch, etc. Complete and provided with a common earth bus for the entire panel and inter connect with the MCCBs as per IE rules and code of practice. The panel shall be fitted on a common base frame on suitable foundation. (Specification may be modified if there is only one) pump set by changing 2 Nos of outgoing as one out going and the provision of interlocking may be avoided).

**5. Cabling work**

Supply, laying, testing and commissioning of suitable size XLPE cables for the above pump set from panel board to starter and from starter to motor. Considering energy conservation.

**6. Earthing**

Supply of all materials and providing suitable earthing by using suitable size pipe/plate earthing GI/copper strip buried in ground and giving double earthing to motor, starter, panel board etc. as per IE standards.

**7. Capacitor**

Supply, erection, testing and commissioning of heavy duty APP capacitor for the motor to get a power factor above 0.95. The capacitor shall confirm to IS 2834.

**8. Valves**

Supply of suitable size best quality heavy duty ISI marked CI double flanged sluice valve and Non return valve with suitable pressure rating. Fitting of valves shall include proper RCC support especially for Non return valves. In case of positive suction pump set sluice

valve should be in suction as well as delivery sides for easy repair works.

**9. Suction and delivery pipe connections**

Supply and fitting of suitable size best quality GI/MS pipe of thickness not less than 8mm for a total length of 10 m. suitable for the above pump set and suitable size foot valve and connecting the suction pipe and valve with suitable flanges, nut and bolts IR sheet etc. complete and connecting the delivery side of the pump with the valves and suitable specials to connect with the pumping main. If the dia of suction pipe is above 80 mm CI pipes & specials shall be provided. Also two year Warrantee and Supply of ELCB/RCCB to be added at panel for safety.

Steel work in built up M.S. Header pipe (suitable size), 8mm thick including cutting, hoisting, fixing in position and applying a priming coat of approved steel primer, including welding etc. complete as per the direction of the dept officers.

**10. Statutory Approval**

Approval from Electrical Inspectorate including preparation of drawings, obtaining approval for the drawings, arranging inspection of the installations and obtaining safety certificate et; after completion of the works. The necessary statutory fees, if any, payable to the Electrical Inspectorate / KSEB will be paid by KWA.

## 5. DESIGN, MANUFACTURING, SUPPLY AND ERECTION OF TRANSFORMER

**A) Design, Manufacturing, Supply and Erection of 630 KVA Transformer and allied works - WTP near Mattupetty Dam**

**5.1 Supply & Erection of suitable internal 11 KV Sub Station and Electrification works near Mattupetty Dam**

5.1.1 Preparation and submission of scheme Plan, Design of Various components and preparation of detailed working Drawings etc and its approval from Electrical Inspectorate.

5.1.2 Design, supply, erection, commissioning and maintenance of 11 KV/433 V Transformer with rating not less than 630 KVA

5.1.3 All electrification works and lighting arrangements.

5.1.4 12 months Maintenance of transformers after the commissioning the work.

The work abstracted in clause **5.1** above includes the completion of works specified in **5.1.5** to **5.1.22** below.

5.1.5 11KV/ 433 V, 630 KVA Cast Resin Type Indoor Transformer with Incomer 11KV VCB panel and metering panel on the incoming side- Supply, erection, testing and commissioning of 630 KVA, 11KV / 433V, 3 Phase, 50 Hz, Self-cooled (Air Natural) Cast Resin dry type, step down double Copper Wound

indoor distribution Transformers having Delta winding on primary and star winding on secondary with Dyn11 vector group connection having neutral brought out on star side with off load tap changer with Circuit tapping + 5 % to - 5 % In steps of 2.5 % with suitable cable box on LT side confirming to IS2026/1977 and IS11771 / 1985 and one 11KV VCB with metering panel attached to the primary side including Manufacturing and Supply of Indoor type 11kV, 630A, 26.3kA for 3 sec Horizontal Draw out VCB fitted with Manual/Motor operated spring charging mechanism with 230V AC motor and motor control switch, shunt trip coil of 24V DC, closing coil of 24V DC, Auxiliary Contact 2NO + 2NC, Mechanical ON/OFF indicator, mechanical operation counter, Protection CT, (--/5A, 5P10, 15VA, 11kV/110V, Cl-0.2, 25VA PT Instantaneous IDMT relay with Microprocessor based 3O/C + and 1EF Protection relay, Strip Heater with thermostat, Digital Ammeter and Digital Voltmeter, Power pack 230V AC-24 V DC emergency OFF push button with HT Metering Panel comprising of 3 Nos of 11KV , 15VA,Class 0.2S , ***/5A (ratio as per contract demand) CT and one no. 11KV/110V ,150VA,Class 0.2S PT unit & Class 0.2S ,-/5A, 3phase 4 wire ,unidirectional ,Static type, TOD Meter with downloading ports as per KSEB specifications suitable for installation in 3 phase , 3wire, 50Hz. AC,11KV, effectively earthed supply system complete as approved by Kerala Electrical Inspectorate and relevant standards.

5.1.6 Supply and laying,testing and commissioning of fallowing size/suitable size XLPE insulated PVC sheathed Armoured Alluminium conductor 11KV cable including escavation,refilling,clamping with clamps partly laid in trench and partly in cable tray,3core 240sq.mm,11KV grade XLPE Aluminium UG cable laying from KSEB pole to incomer VCB

5.1.7 Supply of heat shrinkable type cable end kit with all materials and providing end termination of the following size XLPE insulated, armoured Aluminium conductor cable for giving power connections to HT Breaker, Transformers

etc. complete as per standards.3 core 240 Sq.mm.11kv,XLPE AL Cable.

5.1.8 Main Switch Board (MSB) -Design, engineering, supply, erection, testing and commissioning of medium voltage sheet steel clad, totally enclosed, dust tight & vermin proof, indoor floor mounted cubicle type switchboard fabricated out

of 14 swg mild steel sheet and bus duct of high conductivity aluminium bus bars, PVC sleeved, air insulated and of adequate size as mentioned, for operation on 3 phase, 4 wire, 415 volts, 50 Hz. AC supply system. The ACBs,MCCBs, all meters and relays etc. should be of reputed make and conforming IS specifications and are assembled, interconnected, earthed and the switch board should be painted with anti-rust primer coating followed by two coats of paint of approved shade, by Kerala Electrical Inspectorate and relevant standards. 1000 A TPN bus bars , Earthing conductor : 32 x 3 mm copper consisting of (a)Incomer: 1No 1000A ,50kA 4P EDO microprocessor based ACB with overload, short circuit, instantaneous and earth fault protection and with shunt trip and UV release without any deration up to 50 degree - 1 set CTs: 1000/5 15 VA CL0.5 -4 Nos RYB phase indicating lamps (LED ) with control fuses and switches -1 set. Digital Energy meter with RS 485 port, class 0.5s, b) Outgoings- 200 A 36KA ,TPN MCCB- 3 Nos (One to 85HP motor control Panel ,another for APFC panel(future) and the other to lighting Switch Board) ,100 A 36KA TPN MCCB - 3 Nos (to 40HP motor control Panel and 2 spare for future expansion) , and 500A,36KA,TPN, MCCB - 2Nos (One to 250HP motor control Panel and one spare)re as outgoings Metering compartments comprising of voltmeters, ammeters power factor meters, Selector switches, protection relays as per statutes, energy meters for all outgoing feeders, indicating lamps etc. complete as per KEI Standards. The panel shall be guaranteed for at least one year.

5.1.9 Supply and Laying testing and commissioning of following sizes of (1100V) cable XLPE Aluminium - 3.5C x 400 Sqmm ( Transformer to Main Switch Board).

5.1.10 End terminations 3.5 core 400Sq.mm. 1100V AYFY Cable using Brass Glands and Aluminium Sockets.

5.1.11 Preparation of Electrical drawings, getting approval from the electrical Inspectorate , giving completion report at KSEBL and getting Electrical

connections from KSEB at Well.

5.1.12 Wiring for light point/ fan point/ exhaust fan point/ call bell point with 1.5 sq.mm FRLS PVC insulated copper conductor single core cable in surface / recessed medium class PVC conduit, with modular switch, modular plate,

suitable size GI box and earthing the point with 1.5 sq.mm. FRLS PVC insulated copper conductor single core cable etc as required. Group B.

5.1.13 Wiring to the 5 Amps plug points with 3/22 PVC insulated copper conductor along with single run 3/22 PVC insulated copper conductor (Green colour) drawn inside suitable size PVC conduit pipe with necessary bends, elbows, junction boxes etc. in open conduit system clamped on struss/wall with suitable clamps and saddle clips including supplying and fixing of 5 Amps flush type 3 pin plug control switches fixed in combined position on suitable M.S. box with 3 mm thick hylem sheet cover.<br>(Cost of Materials, Erection, Testing and Commissioning charges).

5.1.14 Supply, fabrication and fixing of MS angle/supports, cable trays for clamping of cables in trenches/ through walls and trusses etc. Angles and plates should be painted one coat of primer and two coat of enamel paint as required as per KEI standards.

5.1.15 Supply and providing plate earthing as per IS 3043 with 1200x1200x12 mm GI/CI earth plate, 50 mm GI watering pipe fixed to the earth plate with 25 x 3mm GI clamps, GI funnel with weld mesh, filling required quantity of charcoal including construction of inspection chamber with a bed concrete of 1:4:8 PCC using 40 mm broken stone (10 cm thick), brick work in cement mortar 1:6, plastering the surface of brick masonry and the exposed surface of PCC bed with cement mortar 1:4, 12 mm thick, (the finished inside dimension shall be 450 x 450 x 450 mm) but excluding test joint, earth continuity conductor to the plate, and covering at the top.

5.1.16 Testing and Conveyance Charges for testing of HT resin cast CTs (4Nos) ,HT PT (1no) and TOD Meter at TMR testing lab of KSEBL.

5.1.17 Payment to be made to KSEB as ECSC (Estimated cost of Service connection) for taking new service connection, Cash Deposit etc.

5.1.18 Supply & laying of one number 3.5C x 25 Sqmm PVC insulated and PVC sheathed armoured aluminium power cable of 1.1KV grade of the following

sizes in ground /Cable Tray.

5.1.19 Supplying and making end termination with brass compression gland and aluminium lugs for following size of PVC insulated and PVC sheathed / XLPE aluminium conductor cable of 1.1 KV grade as required-3.5C x 25sqmm.

5.1.20 Supply & laying the following sizes of Earth Strip for earthing interconnection, termination and leads, for continous earthing along with cables tinned at terminals etc. complete. 25 x 6 mm Copper Strip.

5.1.21 Supply, laying and termination of the following sizes of copper strip/wires for earthing of substation, transformers, HTpanel boards,LT panel boards, equipments and inter connection of earth pits etc., 25X3mm CU strips.

5.1.22 Supply, laying and termination of the following sizes of copper strip/wires for earthing of substation, transformers, HTpanel boards,LT panel boards, equipments and inter connection of earth pits 10SWG Copper.

**STANDARD SPECIFICATIONS**

## 6 LAYING PIPES AND SPECIALS

### 6.1 EARTHWORK EXCAVATION

i) Excavation for the pipe line work is to be done generally in hard soil, medium rock, or ordinary rock. Hard rock is also present along the proposed alignment of the pipe line and excavation in such hard rock may involve stray blasting. Wherever blasting is done for trenching, protection shall be provided with sand bag to void damages to nearby structures. Blasted rock recovered during excavation shall be stacked and handed over to the department.

ii) A 20 cm thick layer of red earth cushion is to be provided in the pipes trenches where the excavated bed of trench is rocky. Excavation shall also be carried out for cutting of the existing pavement formation, where the asphalt surface may have to be cut and removed. Care shall be taken to ensure that damages to the road surface are the least and hindrance to vehicular and pedestrian traffic is minimum. There may be

other utilities lying along the proposed pipe line alignment such as telephone cables, electricity cables, water supply lines etc; and contractor shall have to take maximum care in excavation for laying of the pipe line. If any of the existing structure/utilities are damaged during the excavation for pipe laying works, the contractor shall have to make good the losses/pay the claim of the respective department and Kerala Water Authority will not be responsible for any such expenses incurred by the contractor.

iii) Excavation for laying of pipes shall have to be done in lines so as to provide a minimum cover of 1m above the pipes to be laid and care shall be taken to avoid sudden change in gradient and alignment. The requirement of minimum cover may increase up to 1.20m where the pipes are to be laid over PWD/main highway roads. Also the depth of excavation may increase where the pipe line shall have to cross existing utilities orculvers or drains as decided by the Engineer in charge, based on specific site conditions. Sufficient width as per standard specification and KWA Technical Circular shall be provided. Socket trench shall be taken at every joint before laying pipes so as to enable the body (shell) of the pipe to rest on leveled compacted firm ground in the trench.

iv) Only select fill materials free of stones and sharp aggregates shall be used for refilling the trenches to a height of 30cm from the top of pipeline. Necessary earth shall be supplied by the tenderer without any extra cost if the excavated earth is found insufficient for filling the trenches. All the backfilling shall be watered and compacted in 20cm layers to avoid further settlement of refilled portion. Any excess of soil after refilling the trenches shall be removed by the contractor and disposed off from site within a distance of 5km as directed by the Executive Engineer. If the site condition so warranted, the excavated earth shall be conveyed to a convenient place as directed by the engineer to a dumping place within a distance of 5KM for avoiding inconvenience to normal traffic and the same shall be transported back for

refilling without any extra cost.

v) Protection shall be provided for the filled up earth wherever necessary and as directed by the Engineer in Charge.

vi) In case the contractor fail to provide necessary protection work such as fencing, shoring etc. as required at the site, the Engineer in charge will have the power to deduct the charges of such work at department rates from the payments due to the contractor for the work done by him at such sections. Contractor shall arrange for necessary bailing out water for trenching and pipe laying works and the rate quoted by the tenderer shall be inclusive of all incidental charges such as shoring and any other protection to the sides of trenches. Protections to the telephone and electric post, convenient re-alignment of existing utilities within the proposed alignments, taking all site safety measures such as providing fencing, lighting and watching of the worksite, bailing out water etc. complete should be done by the contractor. No claim for any extra items will be entertained against any of the incidental changes that may arise out of contingencies associated with the completion of the work.

vii) The contractor shall take care to minimize the damages to road surface while trenching is done for pipes laying and construction of valve chambers, anchor blocks etc. Damages to structures like culverts, retaining walls, buildings etc. caused during pipe laying works shall be fully rectified by the contractor at his own cost to the satisfaction of the respective owners of the structures. Compensation if any for such damages shall be met by the contractor alone. Contractor shall use asphalt cutters to minimize damages to tar surfaces. Road crossing portions, the backfilling shall be watered and compacted in 20cm layers to avoid further settlement of refilled portion and shall be provided with concrete 1:2:4 for a thickness of 25 cm in level with the existing road surface.

## 6.2. LAYING PIPES & SPECIALS

The pipe shall be laid in accordance with approved lay out plan. However the Executive Engineer has the authority to change the alignment for better benefits to public and department during execution. In such cases the contractor shall lay pipelines as per direction of the Executive Engineer without any additional cost.

Pipes shall be laid in straight line in trenches so that the body directly rests on the soil bed. Handling of pipes to lay in trenches or in any support structure must be done carefully and proper equipments should be used for the purpose. Any deviation from straight line shall be done using standard bends. In case where standard bends are not available for any particular shape of bend, specials fabricated with MS sheets of thickness not less than 8 mm shall be used with the prior approval of the Engineer in charge. *All culverts along the alignment of the pipe line are to be crossed with GI pipes and the bridge crossing shall be with GI / M.S. pipe,* as per the instructions of Engineer in charge and as per approved standards and specifications.

Pipes shall be laid in trenches as per the standard specifications described in respective IS codes for laying pipes. The contractor shall follow the specifications of the IS codes with latest amendments regarding laying of pipes.

## 6.3. JOINTING

Jointing of the pipes shall be as per the IS specification and should be as per the direction of the departmental officers. Jointing of the DI K9 pipe shall be with Tyton rubber rings/lead joint. The rubber ring shall be of best quality and approved make and shall confirm to the relevant IS Code. The rubber ring shall not be prone to termite or any other such insect attack. Jointing of valves shall be in such a way that the removal

is easy. Use of mechanical joints of reputed make will be appreciated.

### 6.4. ROAD CROSSINGS AND BENDS.

All road crossing shall have to be planned in advance and carried out with the concurrence of the Engineer in charge. As far as possible 125 mm to 80 mm, $90^0$ degree bends should be avoided in the road crossings and road bends etc. Contractors will not be permitted to take angular deviation at pipe joints more than 2/3rd of the maximum angle, provided as "allowable" in the relevant specification / product information.

All necessary sanctions for laying pipes along NH/ through fares and road crossings shall be obtained by the Department and the roads and other structures are to be made back to their original condition after laying of pipes except re-tarring of road surfaces. The road crossing of Panchayath roads should be filled red earth and consolidated in layers after watering and ramming, etc. Then the trench should be filled with cement concrete 1: 2: 4 for a minimum depth of 25cm so that the top surface is in level with existing road surface. The same shall be applicable to the edges of roads where severe damages occurred during trenching.

### 6.5. CROSSING OF BRIDGES AND CULVERTS

Crossing of bridges and culverts shall be done by DI flanged pipes/GI pipes/M.S. pipes as the case may be and should be laid exposed. Where the pipe line crosses over the culvert an air valve shall be provided at the crust to avoid air block. Care shall be taken to avoid $90^0$ bends. Culvert crossings may also have to be done at reaches where the above methods are not possible by carrying the pipe line outside the culvert body and supporting it on RCC pillars but without obstructing the vent way. Protection to the exposed pipes in the form of anchor blocks or casing pipes shall be provided.

The contractor shall consult with the local NH/PWD/local body authorities to

finalize the alignment and method of pipe laying at culvert and bridge crossings. The discretion in this regard shall however-be finalized by the Engineer in charge. The contractor shall assess the actual number of culverts/bridge at site before supplying and laying pipes.

### 6.6. FIXING VALVES

Control valves, scour valves and air valves shall be fitted in pipe lines at appropriate locations as suggested by the Engineer in charge. All scour lines are to be laid with scour discharge line of required size flanged CI Class A/DI pipe with sufficient lead length not less than 2.00 m.

### 6.7. VALVE CHAMBERS

RCC valve chambers shall be constructed for all valves of size 80 mm & above, except for air valves provided on exposed pipes at culvert & bridge crossings. Valve chambers shall be provided with a foundation 20 cm thick (in CC 1:3:6 using 50% 40mm broken stone & 50% 20mm broken stone) and RCC cover slab, thickness be not less than 20 cm (in RCC 1:1.5:3 using 20mm broken stone) capable to take load of vehicular traffic designed for class AA loading. For valves of size up to 125mm, valve chambers of inside dimension 120 cm x 120 cm with wall thickness be not less than 15 cm (in RCC 1:2:4 using 20mm broken stone) shall be provided. For valves of size above 150 mm, the inside dimension shall be 150 cm x 150 cm and wall thickness shall be not less than 15 cm (in RCC 1:2:4 using 20mm broken stone). The height of the valve chamber shall be such that the top of cover slab is in level with the road surface. All the valve chambers shall be neatly plastered inside with CM 1:3 and there shall not be any seepage from bottom or sides. Valve chambers deeper than 1.20 m shall be provided with Cl foot rest on sides. The design of valve chambers shall

be submitted to the Executive Engineer and got approved before commencing construction. Foundation to support the valves in CC 1:1.5:3 shall be provided. The contractor should assess the actual no of valve chambers required as per site inspection.

### 6.8. ANCHOR BLOCKS & SUPPORTS

Anchor blocks in CC 1:2:4 as per the standards and specification of KWA/CPHEEO shall be provided in pipe lines at bends, steep slopes and at places as directed by the Engineers in charge. Typical designs shall be got approved by the Engineer in charge before the actual construction. Clamps, wherever necessary, shall be provided with MS flats of minimum 4 mm thickness and 40 mm width and painted with epoxy paints. When pipe lines are to be laid outside of a culvert or bridge, as insisted by the PWD officials, proper support pillars in RCC 1:1.5:3 shall be provided. The design of such support pillars shall be got approved by the Executive Engineer before the construction

### 6.9. HYDROSTATIC PRESSURE TESTING OF PIPE LINES AT FIELD.

After laying the pipe lines, hydrostatic pressure testing shall be done in reaches of not more than 200 m in length as per the relevant IS specifications to ensure that the pipes and joints are sound enough to withstand the maximum pressure likely to be developed under working conditions. Contractor shall not be allowed to proceed with the pipe laying work leaving more than 400m (i.e.2 reaches) of pipelines untested. The joints of all pipes and specials shall be kept open until the testing is completed satisfactorily. The water required for testing the pipe lines, dummy pipes and test pumps etc. shall be arranged by the contractor with his cost. Hydrostatic pressure testing on pipe lines shall be conducted as per the norms of CPHEEO Manual on Water Supply and Treatment (1999). Failure of test due to any reason is to be rectified and retested without any extra cost. Contractor shall maintain a record of testing of

pipeline which shall be got signed by an officer not below the rank of an Assistant Executive Engineer which shall be produced along with his request for release of security deposit and shall be presented to any inspecting officer above the rank of Assistant Executive Engineer. Positive results observed during pre delivery inspection and the acceptance of the inspection report / result by Kerala Water Authority/Agreement authority will not anyway relieve the contractor of his responsibility on the quality of materials supplied by him at the work site. If the materials supplied at site are found to be of inferior quality, the Executive Engineer in charge will have all the right to ask the tenderer to replace the materials supplied at his cost.

### 6.10. FABRICATION AND WELDING

Any fabrication work for bends and specials shall be with minimum 8 mm thick MS plates. The drawing pertaining to such works shall be got approved by the Engineer in charge before the actual fabrication. The schedule of such shop/site of fabrication shall be intimated to the Engineer in charge so as to enable him to inspect the fabrication and welding works. The contractor shall submit welding procedure to the Engineer along with the data regarding certification, experience and other relevant details of the welders and got approved by the Engineer, before the commencement of any welding work for steel structures or specials. The corrosion protection planned for the fabricated components shall also be reported to the Engineer in charge and got approved. The location of the welding shop would be within the area of work and shall be easily accessible for the inspecting officers.

### 6.11. SUPPLY OF SPECIALS

The tenderer may also supply pipe manufactured as per I.S.O. standard but in such

case the guarantee period shall be 60 months. The DI pipe shall be shop lined, centrifugally with cement mortar inside and coated with metallic zinc with bituminous finishing layer outside, all confirming to IS standards. The M.S. Pipes shall be provided with epoxy coating. The work includes supplying and fixing all D.I and M.S specials required for the work including bends, tees, etc, confirming to relevant IS Codes. The specials shall also be lined with metallic zinc with bituminous finishing layer outside confirming to relevant IS Codes. The M.S. specials shall have epoxy coating. The cost of pipe shall be quoted inclusive of the cost of all specials and jointing materials such as labour charges, hemp, white lead etc. Necessary flanged bends according to the site conditions are to be provided. Care shall be taken to avoid bends $90^0$ in the line. DI specials to suite the respective DI (K9) pipes such as bends, Tees, Mechanical joints, Tapers, Flanged sockets etc shall be supplied as per actual requirements at site after getting approval of Executive Engineer.

### 6.12. VENDOR LIST

Two copies of complete list of Vendors/Manufacturers from whom the valves, specials etc. can be procured with the product specification, Brochure/ Product identification number etc. shall be submitted by the contractor on completion of the work.

Note: - The above points will give only general information of the work to be executed under this contract. The bidders shall meticulously inspect the proposed alignment, survey the required areas, water courses which the pipeline will have to cross, road crossing etc. and assess the items of materials to be supplied and works to be done, for satisfactory completion.

## 6.13. PRE–DELIVERY INSPECTION OF PIPES AND OTHER MATERIALS

The contractor shall supply all the materials including pipes and specials covered under the contract at his own cost. The pipes, specials etc. should be tested at the factory by the third party inspection agencies authorized by Kerala Water Authority in presence of a technical officer of Kerala Water Authority deputed by the agreement authority, for witnessing the pre delivery testing by the third party inspectors. In the case an officer from Kerala water Authority deputed for witnessing the pre delivery inspection of materials at factory, the TA & DA of the officer will be paid by the Authority, if the test results are acceptable and positive. In case if the test results are negative, all such expenses of deputing the Kerala Water Authority officer of the pre delivery inspection shall have to be met by the contractor. Inspection certificate should be produced before the dispatch of materials from the factory and only after written communication/after the verification and acceptance of the inspection report, the materials shall be dispatched to the site. The third party inspection is mandatory for all pipes, specials, valves and all electro mechanical equipments supplied by the contractor. Also ISI marked materials shall alone be accepted, wherever ISI specifications is available for such items. Any dispute between the contractor and the supplier or the inspection agencies shall be a matter of settlement among them and the Kerala Water Authority shall not be a party to such dispute. The lump sum amount quoted shall be inclusive of the cost of all materials and charge of inspection as specified here in.

## 6.14. SAFETY AT SITE

Tenderers alone shall be responsible for any damage to men and materials and that may occur at site due to lapses in safety measures. Tenderers shall provide necessary caution boards and fencing as per the directions/specifications of KWA for trenches during the progress of work. Minimum 3 Nos. caution boards shall be placed

in the approaching side and 2 Nos. at the leaving side at the required distance as per the relevant safety codes. They shall also provide sufficient shoring to the sides of trenches to prevent collapse of the sides of the trenches and consequent damages to the men and materials whenever the depth of trench exceeds 1.50m or while excavating near other civil structures. During night hours, contractors shall provide sufficient lighting and watching at the site. Failure to adhere to these clauses shall attract fine and penalties by way of deductions from the payments to contractors at an amount worked out at department rates

**6.15. REPORTS, DESIGNS AND DRAWINGS**

1. Reports, designs and drawings etc. connected with the work shall be furnished by the contractor for approval, minimum in triplicate copies or as called for by the Engineer in charge.

2. Contractor shall prepare and submit to the approving authority seven additional copies of the approved drawings immediately on receipt of approval for transmission to various field offices.

3. On completion of approval of all items connected with the project, the contractor shall prepare three copies of bound volumes of the approved documents and furnish to the approving authority. The preparation and binding of this document should be in such a manner that during the long period of future reference and recorded life, it shall not get disintegrated nor legibility lost.

**6.16. AS LAID MAPS**

Contractor shall submit five hard copies and soft copies in CD, of the AS laid map of the entire facility he had installed for the project as part of this contract with necessary references to the existing permanent structures. The AS laid map must be prepared progressively with the progress of work and must be presented along with all the running

contract bills. Submission of the AS laid map in complete- for the entire work is mandatory for considering the work as completed.

## 6.17. MATERIALS FOR STRUCTURES

### 6.17.1 SCOPE

6.17.1.1 Materials to be used in the work shall conform to the Specifications mentioned on the Drawings, the requirements laid down in this section and Specifications for relevant items of work covered under these Specifications.

6.17.1.2 If any material, not covered in these specifications, is required to be used in the work, it shall conform to relevant Indian Standards, if there are any, or to the requirements specified by the Engineer.

### 6.17.2 CEMENT

6.17.2.1 Cement to be used in the works shall be any of the following types with the prior approval of the Engineer:

a. Ordinary Portland cement, 33 Grade, conforming to IS: 269-1989 or latest revision.
b. Rapid Hardening Portland Cement, conforming to IS: 8041-1990 or latest revision.
c. Ordinary Portland cement, 43 Grade, conforming to IS: 8112-1989 or lates revision.
d. Ordinary Portland cement, 53 Grade, conforming to IS: 12269-1987 or latest revision.
e. Sulphate Resistant Portland cement, conforming to IS: 12330-1988 or latest revision

Cement conforming to IS: 269-1989 or latest revision shall be used only after ensuring that the minimum required design strength could be achieved without exceeding the maximum permissible cement content of 450kg/cum. of concrete.

### 6.17.3 REINFORCEMENT STEEL

Reinforcement steel bars HYSD / TMT bars conforming to IS: 1786 shall be used. The tenderer shall inform the Engineer in charge prior to start of work, regarding the type of steel to be used by him, in the works and provide the samples of all diameter of bars used at site. Necessary test as directed by Engineer in charge should be carried out at reputed institution as directed by the engineer in charge in the samples to affirm the quality. The cost of all such testing shall be borne by the contractor. Only the bars approved by him shall be used in the work.

### 6.17.4 COARSE AGGREGATES

For plain and reinforced cement concrete (PCC and RCC) or prestressed concrete (PSC) works, coarse aggregate shall consist of clean, hard, strong, dense, non-porous and durable pieces of crushed stone, crushed gravel, natural gravel or a suitable combination thereof or other approved inert material. They shall not consist pieces of disintegrated stones, soft, flaky, elongated particles, salt, alkali, vegetable matter or other deleterious materials in such quantities as to reduce the strength and durability of the concrete, or to attack the steel reinforcement. Coarse aggregate having positive alkali silica reaction shall not be used. All coarse aggregates shall conform to IS: 383-1970 and tests for conformity shall be carried out as per IS: 2386-1963 Parts I to VIII or latest revision.

### 6.17.5 SAND/FINE AGGREGATES

**6.17.5.1** For masonry work, sand shall conform to the requirements of IS: 2116-1980 or latest revision.

**6.17.5.2** For plain and reinforced cement concrete (PCC and RCC) or prestressed concrete (PSC) works, fine aggregate shall consist of clean, hard, strong and durable pieces of crushed stone, crushed gravel, or a suitable combination of natural sand, crushed

stone or gravel. They shall not contain dust, lumps, soft or flaky, materials, mica or other deleterious materials in such quantities as to reduce the strength and durability of the concrete, or to attack the embedded steel. Motorized sand washing machines should be used to remove impurities from sand. Fine aggregate having positive alkali-silica reaction shall not be used. All fine aggregate shall conform to IS: 383-1970 and test for conformity shall be carried out as per IS: 2386-1963 (Part I to VIII) or its latest revision. The Contractor shall submit to the Engineer the entire information indicated in Appendix A of IS: 383-1970 or latest revision. The fineness modules of fine aggregate shall neither be less than 2.0 nor greater than 3.5.

### 6.17.6 WATER

Water used for mixing and curing shall be clean and free from injurious amounts of oils, acids, alkalis, salts, sugar, organic materials or other substances that may be deleterious to concrete or steel. Potable water is generally considered satisfactory for mixing concrete. Mixing and curing with sea water shall not be permitted. The pH value shall not be less than 6

### 6.17.7 TESTS AND STANDARD OF ACCEPTANCE

All materials, even though stored in an approved manner shall be subjected to an acceptance test prior to their immediate use. Independent testing or cement for every consignment shall be done by the Contractor at site in the laboratory approved by the Engineer before use. Any cement with lower quality than those shown in manufacturer's certificate shall be debarred from use. In case of imported cement, the same series of tests shall be carried out before acceptance.

#### 6.17.7.1 TESTING AND APPROVAL OF MATERIAL

The Contractor shall furnish test certificates from the manufacturer/supplier of materials with each batch of material(s) delivered to site.

The Contractor shall set up a field laboratory with necessary equipment for testing of all materials, finished products used in the construction as per requirements of conditions of contract and the relevant Specifications. The testing of all the materials shall be carried out by the Engineer or his representative for which the Contractor shall make all the necessary arrangements and bear the entire cost.

Tests which cannot be carried out in the field laboratory have to be got done at the Contractor's cost at any recognized laboratory/testing establishments approved by the Engineer.

**6.17.7.2 SAMPLING OF MATERIALS**

Samples provided to the Engineer or his representative for their retention is to be in labeled boxes suitable for storage.

Samples required for approval and testing must be supplied well in advance by at least 48 hours or minimum period required for carrying out relevant tests to allow for testing and approval. Delay to works arising from the late submission of samples will not be acceptable as a reason for delay in the completion of the works.

If materials are brought from abroad, the cost of sampling/testing whether in India or abroad shall be borne by the Contractor.

**6.17.8 REJECTION OF MATERIALS NOT CONFORMING TO THE SPECIFICATIONS**

Any stack or batch of material(s) of which sample(s) does not conform to the prescribed tests and quality shall be rejected by the Engineer or his representative and such materials shall be removed from site by the Contractor at his own cost. Such rejected materials shall not be made acceptable by any modifications.

**6.17.9 TESTING AND APPROVAL OF PLANT AND EQUIPMENT**

All Plants and equipment used for preparing, testing and production of materials for incorporation into the permanent works shall be in accordance with manufacturer's Specifications and shall be got approved by the Engineer before use.

## LIST OF CODE OF PRACTICE FOR REFERENCE

| Sl. No. | Description | Standards |
|---|---|---|
| 1 | HDPE Pipe | IS: 4984/1995 |
| 2 | DI Pipes | IS 8329/2000 |
| 3 | PVC Pipe | IS 4985/1988, IS 4985/2000 |
| 4 | G I Pipes | IS 1239 |
| 5 | Sluice Valve | IS 14846/2000 & IS 2685/1971 |
| 6 | Air Valve | IS 4845/2000 |
| 7 | Cement | IS 269/1976 & IS 8112/1989 |
| 8 | Steel Bars | IS 432-Part I, IS 1139 & IS 1786 |
| 9 | Sand | IS 2116/1985 & IS 1542/1977 |
| 10 | Water for mixing & Curing | 4.3 of IS 456/2000 |
| 11 | Coarse & fine aggregates for concrete | IS 383/1970 & IS 456/2000 |
| 12 | Admixture | IS 9103/1972 |
| 13 | DI & PVC Specials | IS 9523/2000 & IS 7834/1987 |
| 14 | Non standard MS bend | IS 816/1969 & IS 822/1970 & IS 823/1964 |
| 15 | Rubber Rings | IS 12820/2004 & IS 5382/1985 |
| 16 | Solvent cement for joints | IS 7634/2003 – Part III |
| 17 | Anti corrosive epoxy painting for MS | IS 6916/1973 |
| 18 | Pipe line pressure test | IS 12288/1987 |
| 19 | Lightning arrestors | IS 4850 |
| 20 | Drinking water standards | IS 10500/1983 |
| 21 | Vaccum type chlorinators | IS 10553/1983 – Part I & II |
| 22 | Steel structures | IS 800 |

| | | |
|---|---|---|
| 23 | Sluice Valve with gear arrangements | IS 780/1984 |
| 24 | Testing of concrete | IS 1199/1959 & IS 516/1959 & IS 456/2000 |
| 25 | Mixing of concrete | Clause 8 & 9 of IS 456/2000 |
| 26 | Code of practice for concrete structures for storage of liquids | IS 3370/2009 – Part I to IV |
| 27 | Centrifugal pump | IS 1520/1980 |
| 28 | Motors | IS 325/1978 & IS 900/1965 & IS 4029/1967 |
| 29 | Trenching | PHED data Item No. 1001, 1002 & 1003 |
| 30 | Non metallic gasket | ASA B 16.21 |
| 31 | Criteria for Earth Quake design of structures | IS 1893 & IS 4326 |
| 32 | Standard for CI double flanged pipe | IS 1536 & IS 1538 |
| 33 | Steel Pipes | IS 3589: 2001 |

**ANNEXURE A**

## LIST OF CHEMICALS, EQUIPMENTS, GLASS WARES AND FURNITURES REQUIRED FOR THE LABORATORY

### 1.LIST OF CHEMICALS- AR GRADE 'MERCK'

| Sl. No. | Item | Quantity |
|---|---|---|
| 1 | OT Solution | 100 gm |
| 2 | Ammonia | 100 gm |
| 3 | Tissue paper | 3 Rolls |
| 4 | Cotton | 2 Rolls |
| 5 | Kraft Paper | 100 Nos |
| 6 | Ordinary Filter Paper | 25 Nos |

**ANNEXURE B**

## LIST OF EQUIPMENTS

| Sl. No. | Description | Qty |
|---|---|---|
| 1 | Refrigerator 200 Liters with Stabilizer | 1 No |
| 2 | Jar Test Apparatus with 4 Jars | 1 No |
| 3 | pH Meter | 1 No |
| 4 | Turbidity Meter | 1 No |
| 5 | Gas Stove with gas Connection | 1 No |
| 6 | Double Distiller (Pyrex / Corning Glass) | 1 No |
| 7 | Bunson Burner | 5 Nos |
| 8 | Chlorine Comparator | 2 Nos |
| 9 | Stop watch | 1 No |
| 10 | Spirit Lamp | 2 Nos |
| 11 | Servo Stabilizer – 2 KVA Electronics | 1 No |

**ANNEXURE C.**

## LIST OF GLASS WARES (BOROSIL MAKE)

| Sl. No. | Description | Qty |
|---|---|---|
| 1 | Nesslers Tube 100 ml | 10 Nos |
| 2 | Nesslers Tube 50 ml | 5 Nos |
| 3 | Conical Flask 250 ml | 5 Nos |
| 4 | Standard Flask 100 ml | 5 Nos |
| 5 | Standard Flask 250 ml | 5 Nos |
| 6 | Standard Flask 500 mls | 5 Nos |
| 7 | Graduated pipette 10 ml | 5 Nos |
| 8 | Burette25 ml | 3 Nos |

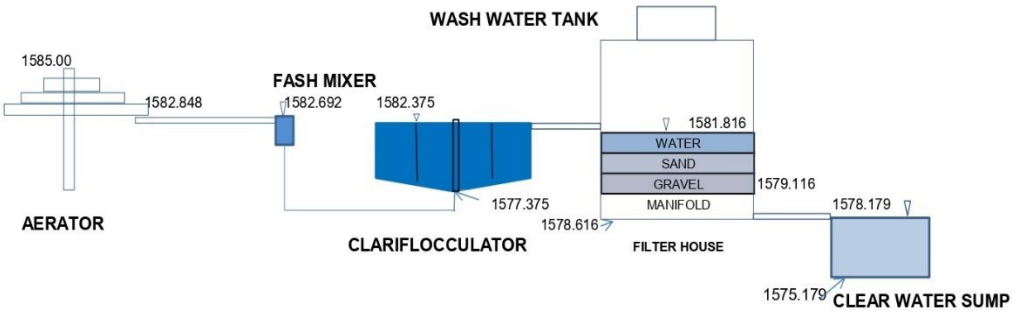
| | | |
|---|---|---|
| 9 | Burette 50 ml | 3 Nos |
| 10 | Burette 100 ml | 2 Nos |
| 11 | Burette Stand | 8 Nos |
| 12 | Nesslers Tube Stand (Fibre made) | 5 Nos |
| 13 | Pipette Stand (Fibre) | 5 Nos |
| 14 | Pipette barrel | 5 Nos |
| 15 | Sampling Bottles 125 ml | 25 Nos |
| 16 | Stoppered Bottles Amber coloured (1 ltr) | 5 Nos |
| 17 | Stoppered Bottles 500 ml | 10 Nos |
| 18 | Stoppered Bottles 1 Lit | 10 Nos |
| 19 | Glass Rod | 1 Dozen |
| 20 | Rubber Bulb with various pours | 5 Nos |

**ANNEXURE D.**

## 1. FURNITURE REQUIREMENT

| Sl No | Item | Quantity |
|---|---|---|
| 1 | Stool (Teak wood) | 4 Nos |
| 2 | Armed Chair | 6 Nos |
| 3 | Office table with drawers on both end and glass slab at the top | 2 Nos |
| 4 | Glass door cupboard. | 1 No |
| 5 | Ordinary Chairs | 4 Nos |
| 6 | Steel Almarah with locker | 1 No |
| 7 | Table, Chair and coat | 1 No each |

**JJM- WSS TO MUNNAR, PALLIVASAL, DEVIKULAM AND CHINNAKANAL PANCHAYATHS IN IDUKKI DISTRICT - Construction of 22 MLD Treatment Plant near Mattupetty Dam**



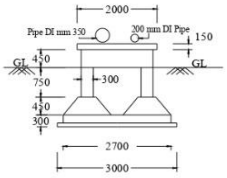

**HYDRAULIC FLOW DIAGRAM**

Total head loss = 6.82 m

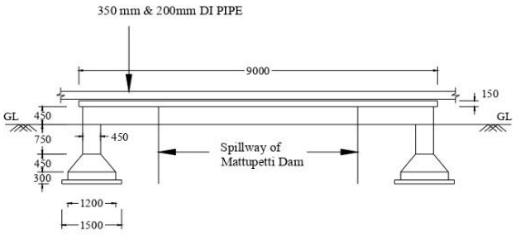
ASSISTANT ENGINEER
PROJECT SECTION, KWA
THODUPUZHA

Assistant Executive Engineer
No. II
Kerala Water Authority
Project Division, Kattappana

**Superintending Engineer**




**CROSS SECTION - BRIDGE FOR SUPPORTING PIPE LAYING**




**LONGITUDINAL SECTION - BRIDGE FOR SUPPORTING PIPE LAYING**

ASSISTANT ENGINEER
PROJECT SECTION, KWA
THODUPUZHA

Assistant Executive Engineer
No. II
Kerala Water Authority
Project Division, Kattappana

KERALA WATER AUTHORITY
Project Division, Kattappana

JJM - WSS TO MUNNAR, PALLIVASAL, DEVIKULAM & CHINNAKANAL PANCHAYATHS IN IDUKKI DISTRICT - DETAILS OF BRIDGE FOR SUPPORTING PIPE LAYING

GENERAL ARRANGEMENT